



3.5.2 Physics Paper 2 (232/2)

SECTION A (25 marks)

Answer **all** the questions in this section.

- 1 State the reason why when a ray of light strikes a mirror at 90° , the reflected ray travels along the same path as the incident ray. (1 mark)
- 2 Explain why the image formed in a pin hole camera gets blurred when the hole is enlarged. (2 marks)
- 3 State the reason why the magnetic field strength of a magnet is greatest at the poles. (1 mark)
- 4 **Figure 1** shows a cell of e.m.f. 2 V connected in series with a resistor R and a switch S. Voltmeters V_1 and V_2 are connected across the cell and the resistor respectively.

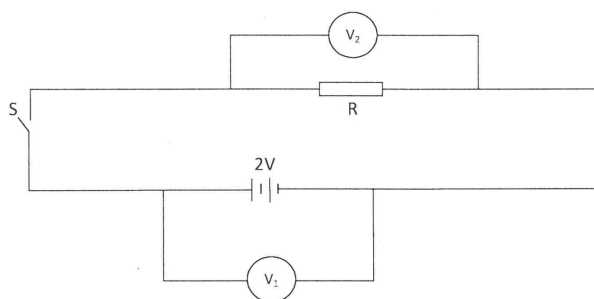
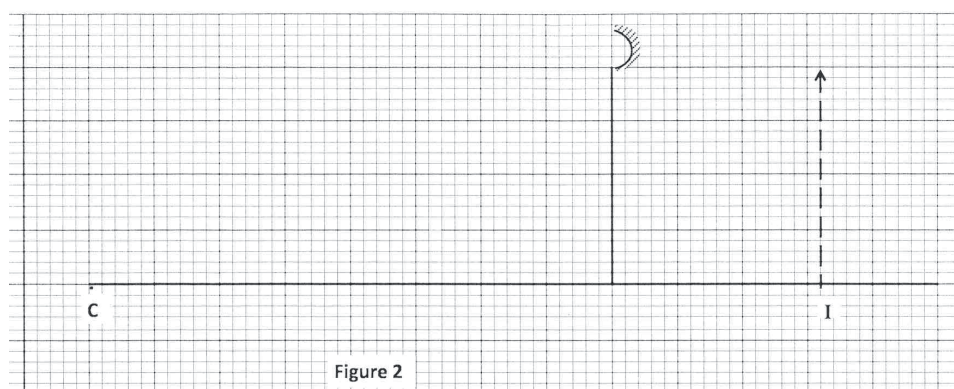


Figure 1

- (a) State the reading of V_1 with S open. (1 mark)
 - (b) With S closed, V_1 reads 1.6 V. State the reading of V_2 . (1 mark)
- 5 **Figure 2** shows the image of an object formed by reflection in a converging mirror. C is the centre of curvature of the mirror.



Complete the diagram to show:

- (a) how incident rays are reflected to form the image; (2 marks)
- (b) the object position. (1 mark)



- 6 **Figure 3** shows a ray of light passing into a glass prism ABC.

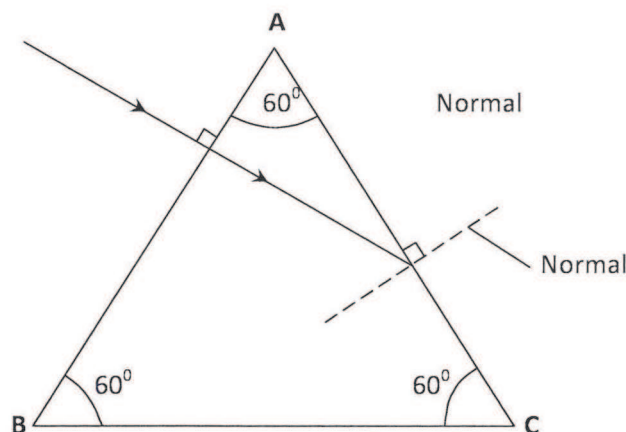
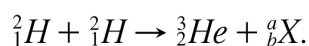


Figure 3

Sketch the path of the ray as it travels from face AC. (*critical angle for glass is 42°*) (2 marks)

- 7 The equation below represents a nuclear reaction in which two deuterium nuclei fuse to form Helium and X.



- (a) Determine the values of a and b. (1 mark)
- (b) Identify X. (1 mark)

- 8 **Figure 4** shows a simple transformer connected to a 12 V a.c. source and an a.c. voltmeter.

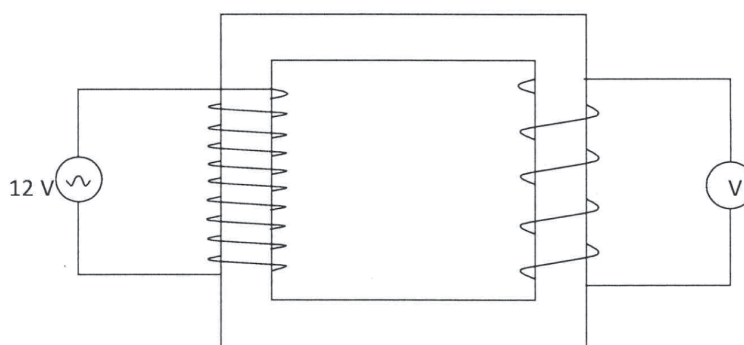


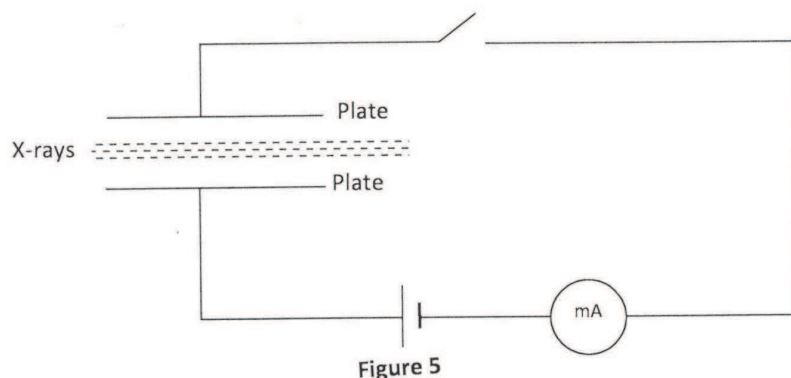
Figure 4

By counting the number of turns in each coil, determine the reading on the voltmeter. (3 marks)

- 9 In domestic wiring systems lamps in the lighting circuit are required to be in parallel and not in series. State **two** reasons for this requirement. (2 marks)



- 10 **Figure 5** shows a narrow beam of x-rays passing between two metal plates in air. The plates are connected in series with a switch, a cell and a milliammeter.



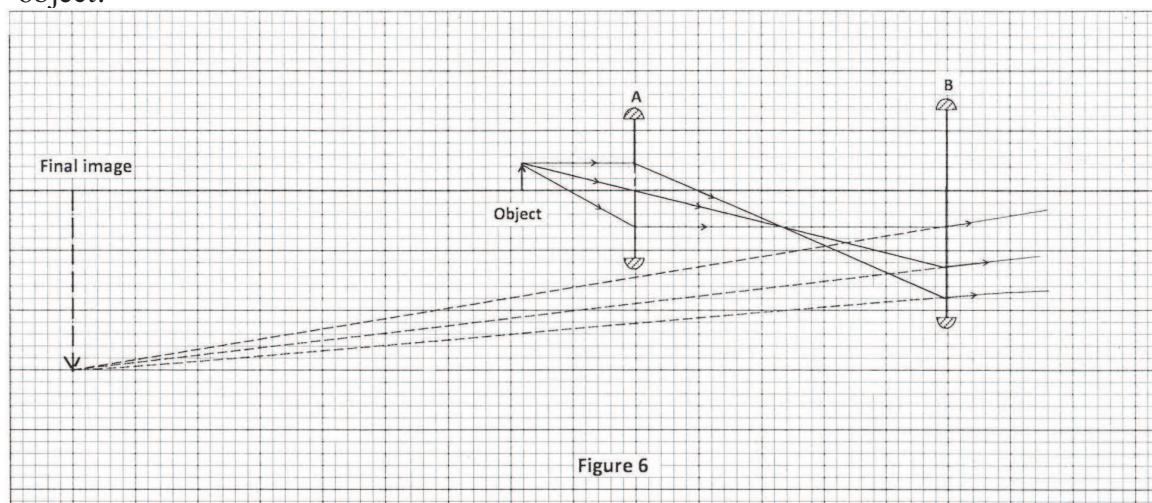
It is observed that when the switch is closed a current flows in the milliammeter. Explain this observation. (2 marks)

- 11 Explain the fact that radiant heat from the sun penetrates a glass sheet while radiant heat from burning wood is cut off by the glass sheet. (2 marks)
- 12 A photon of ultraviolet light having energy E falls on a photoemissive surface whose work function is T . Write an expression for the maximum kinetic energy of the resulting photoelectron in terms of E and T . (1 mark).
- 13 When a germanium crystal is doped with arsenic, it becomes an N-type semiconductor. Explain how this change occurs. (2 marks)
(Number of electrons in the outermost shell for germanium = 4, Arsenic = 5)

SECTION B (55 marks)

Answer **all** the questions in this section.

- 14 **Figure 6** shows two convex lenses A and B used to produce a magnified virtual image of an object.





- (a) Determine the focal length of lens A. (*Take 1 unit to represent 10cm*). (1 mark)
- (b) State the function of:
- (i) lens A (1 mark)
- (ii) lens B (1 mark)
- (c) State how the functions in (b) are achieved by:
- (i) lens A (1 mark)
- (ii) lens B (1 mark)
- (d) Determine the magnification produced by:
- (i) lens A; (2 marks)
- (ii) the whole system. (2 marks)
- 15** (a) Explain how a positively charged electroscope gets discharged when the cap is touched with a finger. (2 marks)
- (b) **Figure 7** shows capacitors **A** and **B** connected in series with a battery of e.m.f 4 V.

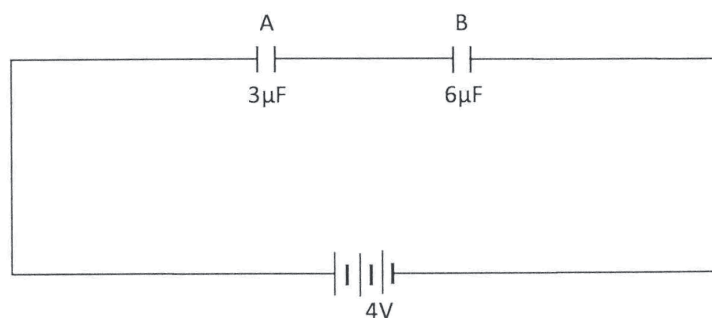


Figure 7

Determine:

- (i) the effective capacitance of the circuit. (3 marks)
- (ii) the quantity of charge in capacitor **A**. (3 marks)
- (iii) the quantity of charge in capacitor **B**. (1 mark)
- (c) **Figure 8** shows an isolated negative point charge **Q**.



Figure 8

On the figure, sketch the electric field pattern around the charge. (2 marks)



- 16 (a) Two points **A** and **B** have a potential difference of **V** volts. **Q** coulombs of charge flow between **A** and **B** for **t** seconds. Determine:
- (i) the electrical energy transformed between the two points in terms of **Q**. (1 mark)
 - (ii) the power transformed in terms of **Q** and **t**. (1 mark)
 - (iii) show that the power transformed is given by $P = IV$. (2 marks)
- (b) The lighting circuit in a house has 20 lamps each rated 60 W, 240 V. Determine whether a fuse rated 4 A can be used in the circuit when all the lamps are put on. (4 marks)
- 17 (a) **Figure 9** shows a cathode ray tube in which a beam of electrons is cast on the screen.

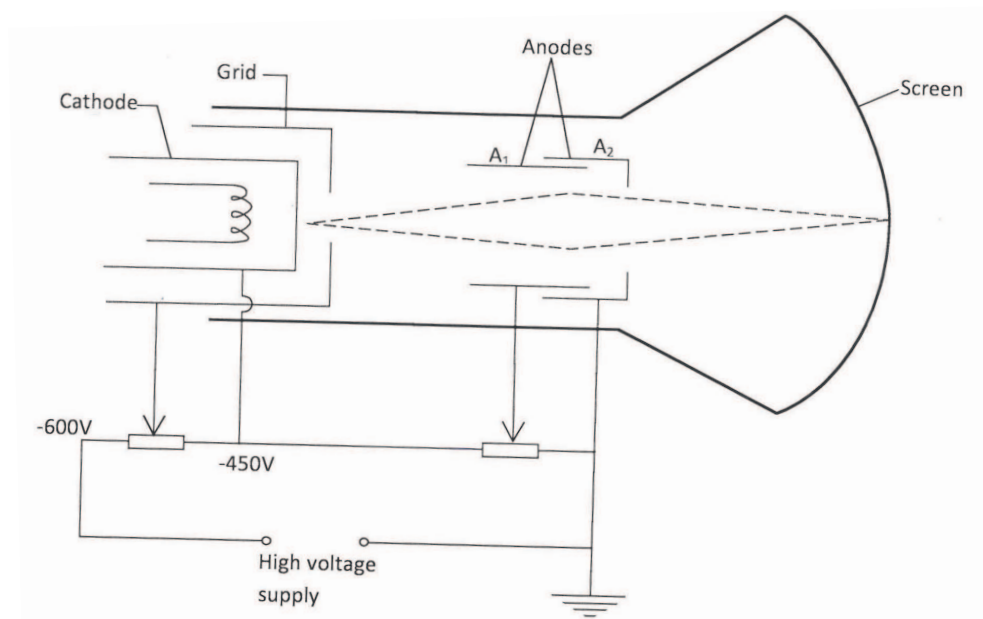
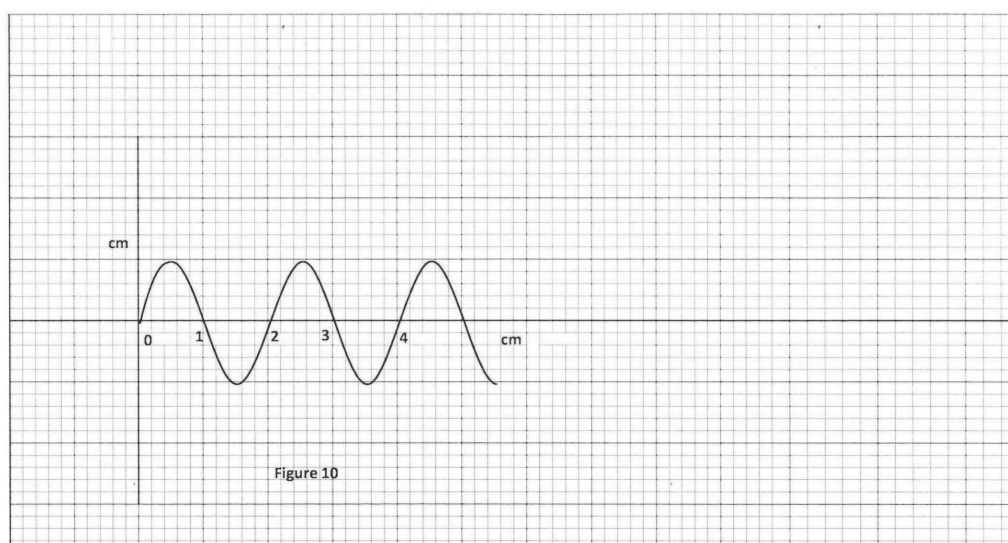


Figure 9

- (i) state how the electrons are produced in the tube. (1 mark)
- (ii) state how the electron beam is detected. (1 mark)
- (iii) State the reason for having a variable potential difference (p.d.) at the:
 - (I) grid; (1 mark)
 - (II) anodes. (1 mark)



- (b) **Figure 10** shows the waveform of a signal applied at the y-plates of an oscilloscope whose time-base is switched to the scale of 2 milliseconds per centimeter.



Determine:

- (i) the period of the signal; (2 marks)
- (ii) the frequency of the signal. (3 marks)

- 18 (a) **Figure 11** shows plane light waves in air incident on a convex lens whose principal focus is **F**, the waves move past point **G**.

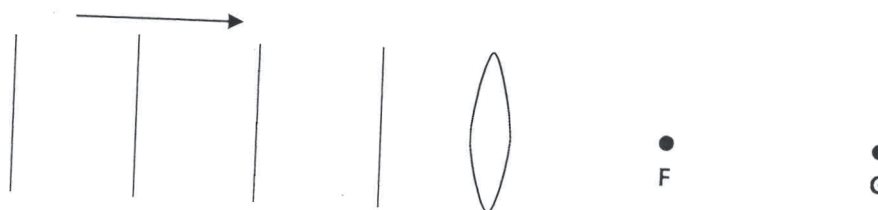


Figure 11

Complete the diagram to show the pattern of the emergent waves between the lens and point **G**. (2 marks)



- (b) **Figure 12** shows crests of circular water waves spreading from two points **A** and **B** due to a vibrator. **C** and **D** are points on the surface of the water.

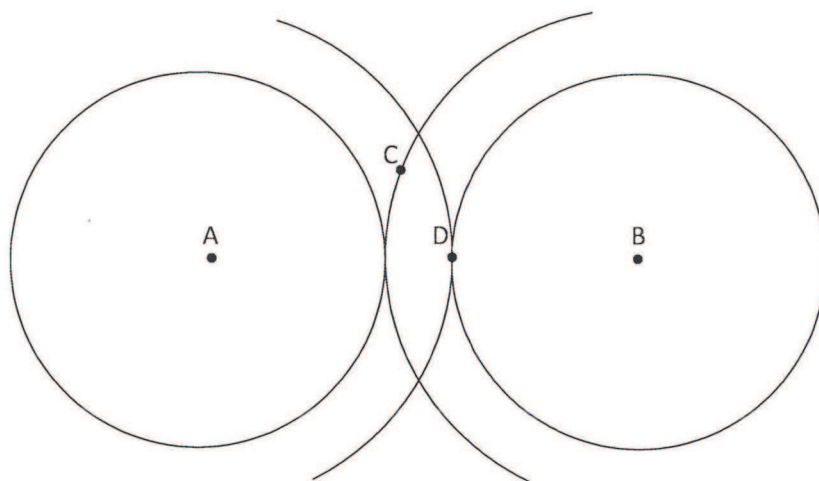


Figure 12

Given that the amplitude of each wave is 5 cm, state with a reason the amplitudes of the waves at point:

- (i) **C**; (2 marks)
- (ii) **D**. (2 marks)
- (c) **Figure 13** shows a standing wave formed when a string of length 1.5 m stretched between two supports is plucked in the middle.

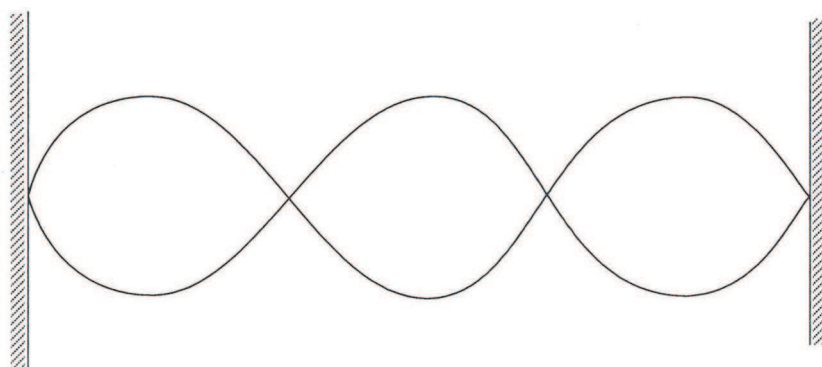


Figure 13

- (i) Explain how the standing wave is formed. (3 marks)
- (ii) Determine the wavelength of the standing wave. (1 mark)



- 19 (a) **Figure 14** shows an E shaped steel block being magnetised by a current through two coils in series.

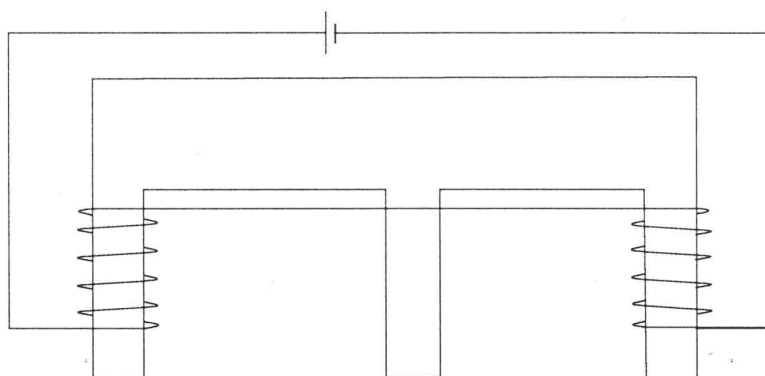


Figure 14

On the figure, indicate

- (i) the north and south poles of the resulting magnet (1 mark)
 - (ii) the complete magnetic field pattern between the poles. (1 mark)
- (b) **Figure 15** shows the permanent magnet made in part (a) above.

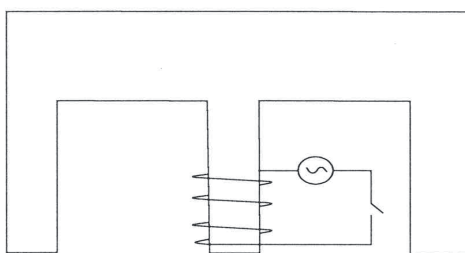


Figure 15

A coil wound loosely on the middle limb is connected in series with a low voltage a.c. and a switch. State and explain the observation made on the coil when the switch is closed. (2 marks)

- (c) In a simple cell, the zinc plate gets negatively charged and the copper plate gets positively charged.
- (i) Name the electrolyte in the cell. (1 mark)
 - (ii) Explain how:
 - (I) Zinc gets negatively charged. (1 mark)
 - (II) Copper gets positively charged (1 mark)
 - (iii) State what constitutes the current when a wire is used to connect the zinc plate and the copper plate externally. (1 mark)